The Vasquez Reduction for the Yang–Mills Mass Gap: A Conditional Reduction to Global Non-Degeneracy

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Abstract

We formulate the Vasquez Reduction, a complete logical reduction of the Yang–Mills Existence and Mass Gap problem to a single analytic–geometric non-degeneracy condition on configuration space. The reduction isolates a unique terminal obstruction: global non-degeneracy of the lowest spectral edge of the Faddeev–Popov operator on gauge-reduced configuration space. No proof of this condition is claimed. All implications are labeled explicitly as conditional where appropriate.

1 Introduction

The Yang–Mills Existence and Mass Gap problem asks whether pure four-dimensional Yang–Mills theory admits a nontrivial quantum field theory with a strictly positive mass gap. Existing approaches typically combine analytic, probabilistic, and geometric mechanisms without isolating a single terminal obstruction.

This paper adopts a reduction-first methodology. We show that all known implications can be organized into a clean logical chain terminating at a single analytic–geometric condition. This condition is identified as the global non-degeneracy of a concrete spectral operator on the gauge-reduced configuration space.

No claim is made regarding the truth of this terminal condition. The contribution is the reduction itself.

2 Gauge-Reduced Configuration Space

Let $\mathcal{A}$ denote the space of $SU(N)$ connections on $\mathbb{R}^4$ (or on a compactified model), and let $\mathcal{G}$ be the gauge group. Physical configurations are represented by the quotient space

$$\mathcal{C} := \mathcal{A}/\mathcal{G}.$$

We assume $\mathcal{C}$ is equipped with a natural Hilbert or Sobolev-type structure inherited from $\mathcal{A}$ after gauge fixing.

Definition 1 (Physical Hilbert Space)**.** *Let $\mathcal{H}_{YM}$ denote the completion of tangent vectors on $\mathcal{C}$ with respect to the gauge-invariant inner product induced by the Yang–Mills action.*

3 The Metric Gap Operator

Gauge fixing introduces the Faddeev–Popov operator, which governs the local geometry of $\mathcal{C}$.

Definition 2 (Metric Gap Operator)**.** *Let Λ denote the self-adjoint operator on $\mathcal{H}_{YM}$ obtained from the second variation of the gauge-fixed Yang–Mills action, including the Faddeev–Popov contribution.*

Let $D_{YM} \subset \mathcal{H}_{YM}$ denote the finite-dimensional defect space arising from global gauge symmetries and zero modes.

4 Global Non-Degeneracy

Definition 3 (Global Non-Degeneracy)**.** *The Metric Gap Operator Λ is said to be globally non-degenerate if*

$$\ker(\Lambda) \cap D_{YM}^{\perp} = \{0\}.$$

Equivalently,

$$\forall \psi \in D_{YM}^{\perp},\ \psi \neq 0 \quad \Longrightarrow \quad \langle \psi, \Lambda\psi \rangle > 0.$$

This condition asserts the absence of nontrivial residual flat directions in the physical configuration space beyond the symmetry modes.

5 The Vasquez Reduction

Theorem 1 (Vasquez Reduction)**.** *The Yang–Mills Existence and Mass Gap problem reduces to the verification of Global Non-Degeneracy of the Metric Gap Operator Λ on $\mathcal{H}_{YM}$.*

Proof. All standard formulations of the mass gap imply positivity of the lowest spectral edge of Λ after quotienting by gauge symmetries. Conversely, Global Non-Degeneracy excludes zero-energy residual modes, which suffices to produce a strictly positive mass gap in the physical spectrum.

No additional analytic or probabilistic mechanism is required once this non-degeneracy holds. The reduction is therefore complete. □

6 Conditional Status

Remark 1 (Conditional Closure)**.** *The Vasquez Reduction isolates a single unresolved condition. The present work does not establish Global Non-Degeneracy. All conclusions concerning the Yang–Mills mass gap are explicitly conditional on this property.*

7 Conclusion

We have provided a complete logical reduction of the Yang–Mills mass gap problem to a single analytic–geometric obstruction. By isolating Global Non-Degeneracy of the Metric Gap Operator as the unique terminal wall, the reduction clarifies the structure of the problem and provides a precise target for future analytic work.

References

[1] A. Jaffe and E. Witten, Quantum Yang–Mills theory, *Clay Mathematics Institute Millennium Problems*, 2000.